

ABSTRACT

In recent years, wearable technology has emerged as a pivotal tool for enhancing personal safety, particularly for women. This paper presents the design and development of a wearable women's safety device integrated with cutting-edge STEM components to detect emergencies and provide immediate alerts.

The system utilizes GPS, GSM, various sensors, a panic button, and an ESP32 microcontroller to ensure rapid response in critical situations. When an emergency is detected, the wearable device captures the user's GPS location and transmits it to a cloud-based monitoring system.

The ESP32 microcontroller plays a dual role, managing real-time data communication and triggering emergency alerts to pre-configured relatives or guardians. The cloud system is developed using c++ offering a user-friendly interface for real-time monitoring of the GPS location. This solution bridges the gap between wearable technology and personal safety, emphasizing reliability, ease of use, and real-time responsiveness. The proposed system showcases the potential of IoT and cloud integration in creating efficient safety solutions for women.

1.INTRODUCTION

Wearable technology has revolutionized the way we interact with devices, offering not only convenience but also significant advancements in personal safety. Particularly for women, who often face unique safety challenges, wearable technology has the potential to be a game-changer in ensuring well-being and providing a rapid response in emergency situations. In recent years, the need for innovative solutions to enhance personal security has driven the development of various wearable devices designed to detect and respond to distress signals in real-time.

One such solution is the wearable women's safety device, a system that integrates multiple STEM components—such as GPS, sensors, a panic button, GSM, and a microcontroller—to deliver immediate alerts and location information in case of an emergency. This paper explores the design and development of such a device, focusing on the integration of the ESP32 microcontroller, which enables the device to communicate efficiently with cloud systems and trigger emergency responses.

The wearable safety device is intended to provide women with a discreet and reliable tool for emergency situations. By capturing GPS data, detecting distress signals, and sending alerts to predefined contacts or guardians, the system ensures that help can be quickly dispatched. Additionally, the use of cloud-based monitoring allows both the wearer and their loved ones to track the device's real-time location.

This system represents the intersection of the Internet of Things (IoT) and personal safety, offering an innovative approach to addressing the challenges women face in today's world. With its emphasis on real-time responsiveness, user-friendliness, and reliability, this wearable safety device has the potential to make a significant impact in enhancing personal security and peace of mind for women everywhere. Through this paper, we aim to highlight the design process, technological integration, and the societal importance of such devices in improving safety standards and empowering women in vulnerable situations.

1.1 OVERVIEW OF THE PROJECT

This project aims to create a wearable safety device specifically designed for women, enhancing personal security through advanced technology. It integrates cutting-edge components like GPS, GSM, sensors, a panic button, and an ESP32 microcontroller to detect emergencies and send immediate alerts. The device constantly monitors the user's location and status, using sensors to detect events like falls, impacts, or other distress signals. When an emergency is detected, the device automatically captures the GPS coordinates of the user and sends them to a cloud-based monitoring system.

The ESP32 microcontroller plays a crucial role in processing the data, ensuring real-time communication between the device and emergency contacts. If the user activates the panic button, the device triggers an alert, sending notifications to pre-configured contacts via SMS or a call. This ensures that help can be immediately summoned, even in remote areas with limited network connectivity.

The cloud system offers a user-friendly interface for emergency contacts to track the user's real-time location and status. The wearable device prioritizes ease of use, comfort, and reliability, ensuring it can be worn daily without inconvenience. With the integration of IoT and cloud technologies, the device is designed to deliver an efficient, scalable safety solution, offering peace of mind to women in potentially dangerous situations.

This innovative wearable device exemplifies how technology can improve personal safety, offering a quick response mechanism in emergencies and making use of modern communication and monitoring systems to enhance protection.

NETWORK SECURITY

1. Data Privacy:

- GPS location data is highly sensitive.
- Potential for unauthorized access to personal information (contacts, emergency alerts).
- Risk of data interception during transmission.

2. Communication Security:

- GSM communication can be intercepted.
- Cloud-based data exchange needs strong encryption.

- Vulnerabilities in the ESP32's Wi-Fi/Bluetooth capabilities.
3. Device Security:
 - Physical tampering with the wearable device.
 - Malicious software injection into the ESP32.
 - Unauthorized access to the panic button or sensor data.
 4. Cloud Security:
 - Unauthorized access to the cloud server.
 - Data breaches on the cloud platform.
 - Denial-of-service attacks on the monitoring system.

NETWORK SECURITY MEASURES:

1. Data Encryption:
 - End-to-End Encryption: Encrypt GPS location and sensor data on the ESP32 before transmission, and decrypt it only on the authorized cloud server or recipient's device. Use strong encryption algorithms (e.g., AES-256).
 - HTTPS/TLS: Use HTTPS for all communication between the ESP32, GSM module, and the cloud server. This ensures that data is encrypted during transit.
 - Encryption(data)→Transmit→Decryption(data)
2. Authentication and Authorization:
 - Strong Authentication: Implement robust user authentication for the cloud-based monitoring system (e.g., multi-factor authentication).
 - Device Authentication: Mutually authenticate the ESP32 and the cloud server to prevent unauthorized devices from connecting.
 - Role-Based Access Control (RBAC): Restrict access to sensitive data based on user roles and permissions. Only authorized personnel (e.g., guardians, emergency services) should have access to location data.
 - API Keys: Use API keys for the ESP32 to communicate with the cloud, and rotate these keys regularly.
3. Secure Communication Protocols:
 - GSM Security: Use GSM security features like A5/1 or A5/3 encryption (if available) to protect cellular communication.
 - Wi-Fi Security (if used): Use WPA2/WPA3 encryption for Wi-Fi communication between the ESP32 and the network.

- Secure MQTT (if used for IOT communication): If MQTT is used for the IOT communication, use TLS encryption, and secure authentication.

4. Device Security:

- Secure Boot: Implement secure boot on the ESP32 to prevent unauthorized firmware from running.
- Firmware Updates: Provide secure over-the-air (OTA) firmware updates to patch vulnerabilities.
- Tamper Detection: Implement hardware or software-based tamper detection to alert users if the device has been physically compromised.
- Secure Storage: Store sensitive data (e.g., encryption keys, user credentials) in secure storage on the ESP32.

5. Cloud Security:

- Firewall: Use a firewall to protect the cloud server from unauthorized access.
- Intrusion Detection/Prevention Systems (IDS/IPS): Implement IDS/IPS to detect and prevent malicious activity.
- Regular Security Audits: Conduct regular security audits and penetration testing to identify and address vulnerabilities.¹
- Data Backup and Recovery: Implement robust data backup and recovery procedures to protect against data loss.
- Secure Coding Practices: Adhere to secure coding practices during the development of the cloud-based system (C++).

6. Privacy by Design:

- Data Minimization: Collect only the necessary data.
- Data Retention Policies: Implement clear data retention policies and delete data when it is no longer needed.
- User Consent: Obtain explicit user consent before collecting and sharing personal data.
- Anonymization/Pseudonymization: Anonymize or pseudonymize data whenever possible.

7. Emergency Response Security:

- Secure Alert Channels: Ensure that emergency alerts are transmitted through secure channels (e.g., encrypted SMS, secure push notifications).
- Verification Mechanisms: Implement verification mechanisms to prevent false alarms.
- Access Control for Emergency Services: Implement strict access control for emergency services to ensure that only authorized personnel can access location data.

By implementing these network security measures, you can significantly enhance the safety and reliability of the wearable women's safety device, protecting sensitive data and ensuring rapid response in critical situations.

1.2 ABOUT THE ORGANIZATION

SUN SOFTWARE SOLUTIONS is a company in India started with business acumen in providing a user friendly feature and customized solutions for small, medium, and large businesses. The company provides full-cycle services in the areas of customized IOT projects ,software development, web designing, web designing, web development, SEO and software solutions and services.

SUN SOFTWARE SOLUTIONS is known for the immense creativity, expertise and tendency to always bring something new to the table. Simply put, if you are looking for a IOT and web development company to help you with an inspiring and inventive project, we should be first in your list.

VISION AND MISSION

The company vision is to “develop in a constant manner and grow as a major IT service provider to become a leading performer, in a providing quality IOT ,web and software development solutions in the competitive global market place.

The company mission is to “Enhancing the business growth of our customers with creative design and development to deliver market-defining high-quality solutions that create value and reliable competitive advantage for our clients around the world”.

2. SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

Existing wearable safety devices primarily focus on providing basic emergency alerts through panic buttons and GPS tracking. These devices typically send distress signals via SMS or phone calls to predefined emergency contacts when activated. Popular examples include safety wearables like Safety Wearable Pendants and SOS smartwatches, which integrate GPS to transmit the wearer's location in case of emergency. Many systems also include mobile applications to allow real-time monitoring of the device. However, these existing solutions often lack advanced features such as environmental sensors, real-time cloud monitoring, or autonomous emergency detection. Additionally, power consumption remains a significant challenge, with many devices requiring frequent charging. While some systems attempt to integrate IoT and cloud technologies, they still face issues with data security, reliability, and scalability. Despite these limitations, existing wearable safety devices have laid the foundation for future, more sophisticated solutions.

DISADVANTAGE

- Frequent charging is required, which can be inconvenient and render the device useless if the battery dies during an emergency.
- Issues with data security, reliability, and scalability persist, particularly in systems attempting to integrate IoT and cloud technologies.
- Although mobile applications enable real-time monitoring, the level of detail and responsiveness may be limited.
- Cloud based solutions can have trouble with scalability when large numbers of users are using the system at the same time.

2.2 PROPOSE SYSTEM

The proposed system is a wearable women's safety device designed to provide real-time emergency alerts and location tracking using advanced STEM components. It integrates an ESP32 microcontroller, GPS module, and a panic button, allowing the wearer to trigger an alert with a simple press. Upon activation, the device captures the user's location and transmits it to a cloud-based monitoring system.

The cloud system is developed with C++ offering a user-friendly interface for real-time tracking. Additionally, the system features low power consumption, ensuring long battery life for continuous use. By combining IoT and cloud integration, the device ensures fast and reliable responses in critical situations. The system aims to bridge the gap between wearable technology and personal safety with a focus on real-time responsiveness and ease of use. It also provides a scalable solution for expanding safety features, such as adding biometric or environmental sensors.

ADVANTAGE

- Provides immediate alerts to predefined emergency contacts, ensuring swift assistance.
- Accurate GPS location tracking enables precise localization of the wearer in distress.
- Simple panic button activation allows for easy triggering of emergency alerts.
- Designed for extended battery life, ensuring continuous operation and reliability in emergencies.
- Leverages IoT and cloud technologies for seamless data transmission and real-time monitoring.
- Facilitates fast and reliable responses in critical situations.
- The system is designed to provide fast and reliable responses.
- The system helps to bridge the gap between wearable technology, and personal safety.

3. SYSTEM REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

1. NODE MCU - ESP32 Development Board
2. GPS MODULE - Supplies the critical location data that is transmitted to emergency contacts and the cloud monitoring system.
3. TACTILE PUSH BUTTON - Allows the wearer to manually trigger an emergency alert by pressing the button.
4. IOT - IoT communication protocols (e.g., MQTT, HTTP)
5. GSM - (Global System for Mobile Communications) Module (SIM800L, SIM900)

3.2 SOFTWARE REQUIREMENTS

1. WINDOWS - USB ports for connecting the Arduino or ESP32 board.
2. ARDUINO IDE - The Arduino IDE is used to write, compile, and upload code (sketches) to Arduino and ESP32 microcontrollers.
3. BLYNK - Blynk cloud server.

4. SYSTEM OVERVIEW

4.1 MODULE DESCRIPTION

TACTILE PUSH BUTTON:

A tactile push button is a type of switch that is designed to be activated by applying physical pressure to it, typically by pressing it with a finger or some other object. These buttons are commonly used in various electronic devices and control panels where user input is required. They provide a tactile feedback, which means that when the button is pressed, you can feel a physical "click" or resistance, confirming that the button press has been registered. Tactile push buttons usually consist of a button cap, a plunger or actuator, and a switch mechanism underneath. The switch mechanism is responsible for making or breaking an electrical connection when the button is pressed or released. The tactile feedback is often achieved through the use of a dome switch, which is a small metal dome that collapses and then rebounds when the button is pressed, producing the characteristic tactile sensation and audible click.



Fig. Tactile Push Button

Pin Description

1. **Common Terminal (COM):** This pin is usually the common connection for the switch and is connected to one side of the switch contact.

2. **Open Terminal (NO):** When the switch is not pressed, this terminal is not connected to the common terminal. When the switch is pressed, it makes a connection to the common terminal.
3. **Normally Closed Terminal (NC):** When the switch is not pressed, this terminal is connected to the common terminal. When the switch is pressed, it breaks the connection to the common terminal.
4. **Ground (GND):** Some tactile switches might have a separate pin for grounding purposes, especially if they are used in electronic circuits.

Specifications:

Operating Voltage	12V DC
Contact Configuration	Normally Open (NO)
Contact Rating	100mA
Actuation Force	160g
Travel Distance	0.3mm
Life Cycle / Durability	100,000 cycles
Tactile Feedback	Medium
Operating Temperature	-25°C to +70°C
Contact Resistance	< 50 mΩ
Insulation Resistance	> 100 MΩ
Bounce Time	5ms
Mounting Type	Through-hole
Terminal Type	PCB pins
Dimensions	12x12mm

Working

The working principle of a tactile switch involves the use of a spring-loaded mechanism to create a temporary electrical connection when the switch is pressed. Tactile switches are designed to provide tactile feedback to the user, allowing them to feel when the switch is actuated. Here's how the working of a typical tactile switch can be explained:

- **Internal Components:** A tactile switch consists of several internal components, including a housing, a plunger or button, a spring, and electrical contacts.
- **Resting State:** In the resting state (not pressed), the spring exerts pressure on the plunger, keeping it in an extended position. The plunger is located over the electrical contacts. There are typically two sets of contacts inside the switch: Normally Open (NO) and Normally Closed (NC).
- **User Presses the Switch:** When a user applies force to the button or plunger of the tactile switch, it starts compressing the spring. As the spring is compressed, the plunger moves downward.
- **Tactile Feedback:** As the plunger moves downward, the user feels a tactile "click" or resistance. This tactile feedback indicates that the switch has been actuated. The tactile feedback is achieved through the mechanical design of the switch, including the shape of the plunger, the spring's tension, and the housing.
- **Electrical Connection:** As the plunger moves downward, it eventually reaches a point where it pushes the internal electrical contacts together. If the switch is of the Normally Open (NO) type, this action closes the circuit between the common (COM) and NO pins, allowing current to flow through the switch. If the switch is of the Normally Closed (NC) type, pressing the switch opens the circuit between the common (COM) and NC pins, interrupting the current flow.
- **Actuated State:** The switch remains in the actuated state as long as the user continues to apply pressure on the plunger. The electrical connection remains closed (or open, depending on the switch type) during this time.
- **Switch Release:** When the user releases the pressure on the plunger, the compressed spring pushes the plunger back to its extended position. As the plunger moves upward, it separates the electrical contacts. This action returns the switch to its resting state, and the circuit is either opened or closed, depending on the switch type.

A tactile switch uses a combination of mechanical components, such as a spring-loaded plunger and internal contacts, to provide tactile feedback to the user and create a temporary electrical connection when the switch is pressed. The tactile "click" sensation and the electrical behaviour make these switches suitable for various applications where user interaction and momentary switching are required.

GSM

GSM (Global System for Mobile Communications) is a standard developed in the late 1980s for digital cellular networks. It's a foundational technology that has enabled mobile phone communication across the globe. Here's a breakdown:

- **Cellular Networks:** GSM operates through a network of cell towers that divide geographical areas into "cells." When a device communicates, it connects to the nearest tower.
- **SIM Cards:** GSM devices use Subscriber Identity Module (SIM) cards, which contain user information and authenticate the device on the network.
- **Voice and Data:** GSM supports both voice calls and data transmission (like SMS text messages).

Uses of GSM Connectivity

- **Emergency Communication:**
 - It provides a reliable backup communication channel, especially in areas where Wi-Fi is unavailable.
 - The device can send SMS text messages to emergency contacts with location coordinates.
 - It can also enable direct voice calls to emergency services.
- **Location Tracking:**
 - In situations where GPS data needs to be delivered, but wifi is not available, GSM data can be used to transmit the location data.
- **Remote Alerts:**

- GSM can be used to send alerts even when the user is far from wifi networks.



Fig : GSM

Node MCU:

NodeMCU is an open-source firmware and development kit based on the ESP8266 WiFi module. The ESP8266 is a low-cost, highly-integrated wireless microcontroller that gained significant popularity for its ability to provide WiFi connectivity to various electronics projects. The NodeMCU project aims to make it easier for developers and hobbyists to work with the ESP8266 module by providing an easy-to-use firmware and development environment.

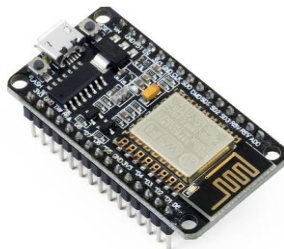


Fig. NodeMCU

Lua Scripting: NodeMCU originally provided a Lua-based scripting environment, allowing developers to write code directly on the module using the Lua programming language. This made it accessible to those who were not familiar with embedded programming.

WiFi Connectivity: The main purpose of the NodeMCU firmware is to enable WiFi connectivity for IoT (Internet of Things) applications. The module can connect to local WiFi networks and communicate with other devices over the internet.

Arduino Compatibility: While the original NodeMCU firmware was based on Lua, there are also Arduino-compatible firmware options available for the ESP8266.

This allows developers to program the module using the Arduino IDE, which is a popular platform for creating embedded projects.

GPIO Pins: The ESP8266 module has a set of General Purpose Input/Output (GPIO) pins that allow you to interface with external components such as sensors, actuators, LEDs, and more.

Integrated Development Environment (IDE): NodeMCU development can be done using various IDEs, such as the Arduino IDE or the NodeMCU-specific IDE. These IDEs provide tools for writing, compiling, and uploading code to the module.

Community and Documentation: NodeMCU has a vibrant and active community that provides tutorials, documentation, and support for users. This makes it easier for beginners to get started with the technology.

Prototyping and Rapid Development: NodeMCU and the ESP8266 are popular choices for rapid prototyping and development of IoT projects due to their low cost and ease of use.

Pin Description

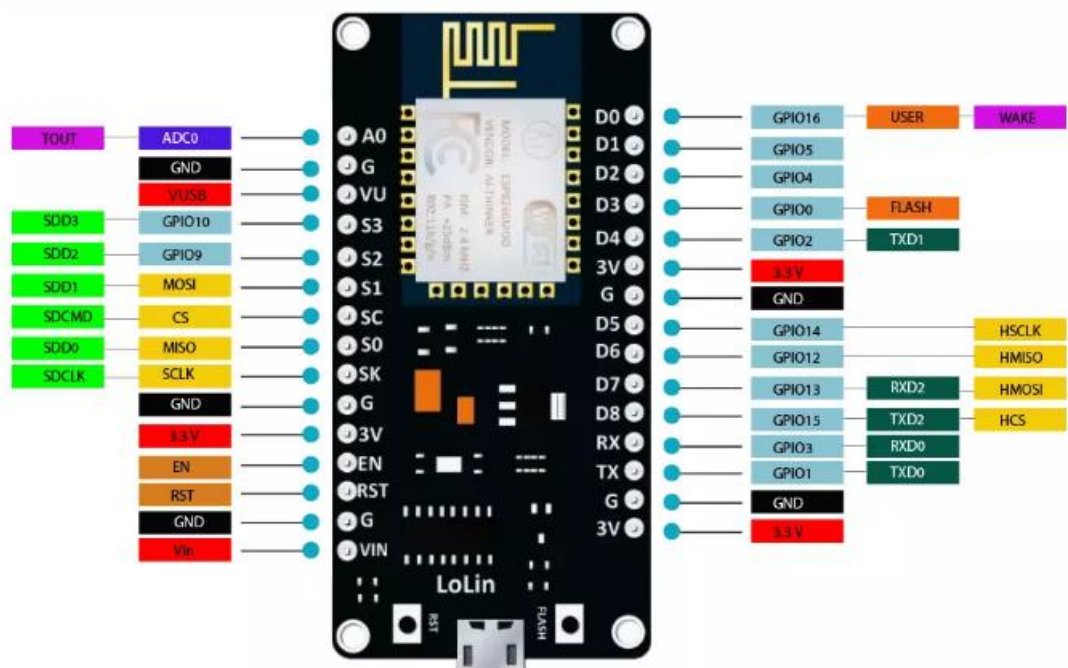


Fig : Pin Description

3V3 (3.3V): Supplies a regulated 3.3V power source for external components. Many components on the board operate at this voltage level.

RST (Reset): Allows you to trigger a reset of the ESP8266, restarting your program or resetting the module.

GND (Ground): Provides the common ground reference for the circuit. Connect components' ground connections here.

D0 (GPIO16): General-purpose digital I/O pin, usable for input or output tasks. Can also wake the ESP8266 from deep sleep.

D1 (GPIO5, SCL): Serves as the clock (SCL) pin for I2C communication, a two-wire serial communication protocol used to connect sensors and devices.

D2 (GPIO4, SDA): Acts as the data (SDA) pin for I2C communication, facilitating the exchange of data between devices.

D3 (GPIO0): General-purpose digital I/O pin. During boot-up, it influences the boot mode of the ESP8266.

D4 (GPIO2): General-purpose digital I/O pin. Also affects boot mode during the boot-up process.

D5 (GPIO14, SCLK): Clock (SCLK) pin used in SPI communication for synchronizing data transfer between devices. It also supports PWM.

D6 (GPIO12, MISO): Master In Slave Out (MISO) pin used in SPI communication for data transmission from the slave to the master. It also supports PWM.

D7 (GPIO13, MOSI): Master Out Slave In (MOSI) pin used in SPI communication for data transmission from the master to the slave.

D8 (GPIO15): General-purpose digital I/O pin. It supports PWM and can be used as an input or output.

TX (GPIO1): Transmit pin for UART serial communication, allowing data to be sent from the board to other devices.

A0 (ADC): Analog input pin for reading analog voltages using the ADC (Analog to Digital Converter). Useful for reading sensors that provide analog output.

Vin: Input voltage pin for external power supply. Typically 5V, used to power the board externally.

EN (Enable): Used to enable or disable the ESP8266 module.

On the NodeMCU v1.0 board, several pins offer Pulse Width Modulation (PWM) capabilities, providing a means to finely control the intensity and behavior of various components in your projects. Pin D1 (GPIO5), D2 (GPIO4), D3 (GPIO0), D5 (GPIO14), D6 (GPIO12), and D7 (GPIO13) all support PWM. This feature allows you to adjust the duty cycle of the output signal, enabling precise control over devices such as LEDs, motors, and servos. By varying the on-off ratio of the PWM signal, you can effectively regulate brightness, speed, and position, making these pins essential for crafting dynamic and responsive projects that require varying levels of output. Remember to consult the documentation and appropriate programming libraries to utilize these pins effectively for PWM-based applications.

Specifications

Microcontroller	ESP8266E
Operating Voltage	3.3V
Digital I/O Pins	11 (D0 - D10)
Analog Input Pins	1 (A0)
Clock Speed	80 MHz
Flash Memory	4 MB
Wi-Fi	802.11 b/g/n (2.4 GHz)
Wireless Range	indoor 100 meter
Outdoor	400 meter
Operating Temperature Range	-40°C to +125°C

Input Voltage	5V (via USB or VIN pin)
Output Voltage	3.3V
Current Consumption	~80 mA (average), ~170 mA (peak)
USB-to-Serial Chip	CH340G
Programming Interface	Micro USB
GPIO Pins	PWM, I2C, SPI, and 1-Wire
Onboard Antenna	Yes
Dimensions	49.3mm x 25.5mm
Compatible IDEs	Arduino IDE, PlatformIO, NodeMCU Lua
CPU	Tensilica L106 32-bit

Working

The NodeMCU v1.0 is a development board that utilizes the ESP8266 microcontroller module, allowing for WiFi connectivity and versatile digital and analog input/output capabilities. Here's a general overview of how the NodeMCU v1.0 works:

- **Microcontroller and CPU:** The NodeMCU v1.0 is centered around the ESP8266EX microcontroller, which features a 32-bit RISC processor. This processor executes instructions, handles tasks, and manages I/O operations.
- **Voltage Regulation:** The board's operating voltage is 3.3V, regulated by an onboard voltage regulator. This ensures that the components receive a stable voltage level for reliable operation.
- **Digital and Analog I/O:** The NodeMCU provides 11 digital I/O pins (D0 - D10) for interacting with the digital world. These pins can be used as inputs or outputs to interface with various devices. Additionally, the A0 pin serves as an analog input with a 10-bit ADC, allowing you to measure continuous voltage levels from sensors.
- **Wi-Fi Connectivity:** One of the standout features of the NodeMCU is its WiFi connectivity. The ESP8266 module supports a range of WiFi modes, including

Station mode for connecting to existing networks and SoftAP mode for creating its own access point. This enables the board to communicate over the internet and with other WiFi-enabled devices.

- **Programming and Communication:** To program the NodeMCU, you can use different Integrated Development Environments (IDEs) like the Arduino IDE or the NodeMCU firmware with Lua scripting. The USB-to-Serial chip (CH340G) facilitates the connection between your computer and the board, allowing you to upload code, monitor output, and debug.
- **GPIO and Communication Protocols:** The General Purpose Input/Output (GPIO) pins are versatile and can be configured for various communication protocols. PWM allows you to modulate the duty cycle of digital signals, I2C lets you connect multiple devices with just a few wires, SPI enables high-speed communication, and 1-Wire simplifies data exchange with sensors.
- **Antenna and Wireless Range:** The NodeMCU v1.0 integrates a PCB antenna that enables wireless communication. The range of the wireless signal depends on environmental factors, signal interference, and antenna design. Generally, it can cover distances of up to 100 meters indoors and up to 400 meters outdoors.
- **Operating System and Applications:** The NodeMCU runs firmware that provides a way to execute your code. This firmware can be written in various languages like C++, Lua, or MicroPython. This versatility opens the door to a multitude of applications, including IoT projects like weather stations, home automation systems, smart appliances, and remote monitoring devices.
- **Power Consumption:** The NodeMCU has varying power consumption levels depending on its operational state. It's important to manage power effectively, especially when running on battery power. You can implement strategies like using deep sleep modes to reduce energy consumption during idle periods.
- **Libraries and Community:** The NodeMCU has a vibrant community that develops and shares libraries, code snippets, and tutorials. This ecosystem can significantly speed up your development process by providing pre-built functions and solutions to common challenges.

NodeMCU v1.0 serves as a powerful platform for creating projects that leverage WiFi connectivity and interact with both digital and analog devices.

Its combination of microcontroller capabilities, communication protocols, and ease of programming make it an excellent choice for both beginners and experienced developers working on IoT, automation, and wireless communication projects.

GPS Module:

A GPS (Global Positioning System) module is a device that allows you to receive signals from GPS satellites and determine your precise location on Earth.



Fig. GPS Module

Satellite Communication: GPS satellites orbit the Earth and continuously transmit signals that include information about the satellite's position and the current time. A GPS module receives these signals and uses the information to calculate its own position.

Triangulation: The GPS module receives signals from multiple satellites. By analyzing the time it takes for the signals to travel from the satellites to the module, the module can calculate the distance to each satellite. With distances from at least four satellites, the module can triangulate its precise location using trilateration algorithms.

Position and Timing: In addition to determining its own latitude, longitude, and altitude, a GPS module can also provide accurate timing information. This timing data is often used in applications that require precise synchronization.

Communication Interfaces: GPS modules typically have communication interfaces that allow them to connect to external devices such as microcontrollers, computers, or smartphones. Common communication protocols include UART (serial communication) and I2C.

NMEA Sentences: Like the NEO-6M module mentioned earlier, GPS modules commonly output data in the form of NMEA sentences.

These sentences contain information about the GPS fix status, latitude, longitude, altitude, speed, and more. NMEA sentences are ASCII strings that can be parsed easily.

Antenna: GPS modules require an external antenna to receive signals from satellites. The quality and placement of the antenna can impact the accuracy of the module's positioning.

Pin Description

VCC: This pin is used to provide power to the GPS module. It's usually connected to a positive supply voltage, such as 3.3V or 5V, depending on the module's specifications.

GND or Ground: This pin is connected to the ground or common reference of the power supply.

TX or Transmit Data: This pin is used for serial communication and sends data from the GPS module to an external device, like a microcontroller or computer.

RX or Receive Data: This pin is also used for serial communication and receives data from an external device.

Specifications

Receiver Type	50-channel u-blox 6 receiver
GPS Bands	L1 C/A code, SBAS, QZSS
Sensitivity	Tracking & Navigation: -161 dBm
Cold Start: -147 dBm	Hot Start: -156 dBm
Position Accuracy CEP	Autonomous: < 2.5 meters CESBAS: < 2.0 meters
Update Rate	Up to 5 Hz (5 times per second)
Protocol Support	NMEA-0183 / u-blox(UBX)
Interface	UART

Antenna	External
Supply Voltage	3.3V
Power Consumption	30mA
Backup Power Consumption	15 μ A
Operating Temperature	-40°C to +85°C
Dimensions	22.4 mm x 17 mm
Positioning Modes	Auto 2D/3D/2Donly/3Donly
Dynamics	Up to 4g acceleration, up to 515 m/s ² velocity
Time Pulse	1 pulse per second (PPS) output

Working

The working principle of a GPS module, such as the NEO-6M, involves receiving signals from multiple GPS satellites orbiting the Earth and using those signals to calculate the module's precise location. Here's how it works in more detail:

- **Satellite Signals:** GPS satellites continuously transmit signals that include information about the satellite's location and the current time. These signals travel at the speed of light.
- **Time of Flight:** The GPS module receives these signals, and by measuring the time it takes for the signals to travel from the satellites to the module, it can determine the distance between itself and each satellite. This is possible because the signal travels at a known speed (the speed of light).
- **Trilateration:** To determine its position, the GPS module needs signals from at least four satellites. With signals from four or more satellites, the module can use a process called trilateration to calculate its exact position. Trilateration involves intersecting spheres centered at the satellite positions, and the intersection point is the GPS receiver's location.

- **Data Processing:** The GPS module processes the received signals to extract information about the satellites' positions, the time the signals were transmitted, and other relevant data.
- **Coordinate Calculation:** Using the time-of-flight measurements and the known positions of the satellites, the GPS module calculates its latitude, longitude, and altitude. This information provides the module's precise position on Earth.
- **NMEA Sentences:** The GPS module typically outputs data in the form of NMEA (National Marine Electronics Association) sentences. These sentences contain information about the GPS fix status, latitude, longitude, altitude, speed, and more. The module transmits these sentences over a serial communication interface (UART) to an external device (like a microcontroller) for further processing.
- **Aiding Data:** Some GPS modules can use aiding data, such as ephemeris and almanac data, to speed up the position calculation process. This data provides information about satellite orbits and positions, helping the module acquire a position fix more quickly.
- **Accuracy and Fix Types:** The accuracy of the GPS module's position calculation depends on factors like the number of satellites in view, signal quality, and the module's processing capabilities. The module can provide different types of fixes, such as "No Fix" (when it can't determine a position), "2D Fix" (latitude and longitude), and "3D Fix" (latitude, longitude, and altitude).
- **Power Management:** To optimize power usage, some GPS modules have power-saving modes. For example, they might go into a low-power mode when not actively acquiring satellite signals.

GPS module works by using precise timing measurements from multiple satellites to calculate its position on Earth. The accuracy of the position fix depends on various factors, including the quality of the signals, the number of satellites in view, and the module's processing capabilities.

4.2 SOFTWARE DESCRIPTION

C++

C++ plays a significant role in the development of Internet of Things (IoT) applications, particularly in scenarios where performance, efficiency, and low-level hardware interaction are crucial. Here's a breakdown of its uses:



Fig: C++

Key Uses of C++ in IoT:

- Embedded Systems Programming:

IoT devices often rely on embedded systems, which are specialized computer systems designed for specific tasks. C++ is highly suitable for programming these systems due to its ability to provide direct hardware control and optimize resource usage.

This is especially important in resource-constrained environments where memory and processing power are limited.

- Real-Time Operating Systems (RTOS):

Many IoT applications require real-time processing, where data must be processed and responded to within strict time constraints. C++ is frequently used in conjunction with RTOS to ensure timely and deterministic responses.

- Network Programming:

IoT devices need to communicate with each other and with cloud-based systems.

C++ provides powerful libraries and APIs for network programming, enabling the implementation of various communication protocols like MQTT, CoAP, and HTTP.

- Data Processing and Analytics:

IoT devices generate vast amounts of data. C++ can be used to perform data processing and analysis directly on the device (edge computing), reducing the need for constant cloud connectivity and minimizing latency.

- Security:

Security is paramount in IoT applications. C++ allows developers to implement robust security measures, such as encryption, authentication, and secure boot processes, to protect devices from unauthorized access and data breaches.

- Hardware interaction:

C++ allows for very low level hardware interaction, which is very important when dealing with sensors, and other hardware components.

- Efficiency:

C++ is a very efficient language, this is very important in IOT devices that are battery powered, or have limited processing power.

Why C++ is Well-Suited for IoT:

- Performance: C++ is known for its high performance, which is essential for real-time applications and resource-constrained devices.
- Low-Level Control: It provides direct access to hardware resources, allowing developers to optimize code for specific devices and applications.
- Portability: C++ code can be ported across different hardware platforms and operating systems, which is important for developing scalable and flexible IoT solutions.

In essence, C++ empowers developers to create efficient, reliable, and secure IoT applications that can handle the demands of real-time data processing and hardware interaction.

Integration of IoT and Cloud-Based Systems:

Recent advancements have integrated cloud technology into wearable safety devices, enabling real-time monitoring and remote assistance. Fauzi et al. (2019) developed an IoT-based personal safety system that used a wearable device connected to a cloud platform. The device could send GPS coordinates to a centralized cloud server, where data could be monitored by family members or authorities. This type of system offers real-time tracking, helping first responders locate individuals in emergency situations more quickly.

Similarly, Kumar et al. (2020) explored the use of cloud-based systems for wearables that combine GPS data with a mobile app interface. Their system, designed for women's safety, could send alerts, track real-time location, and even record voice or ambient noise in the case of emergencies. The cloud platform in their design allowed emergency responders to directly access the data and make informed decisions.

5.SYSTEM DESIGN

5.1 ARCHITECTURE DESIGN CONCEPTS

1. Emergency Detection:

- **Panic Button:** A crucial, easily accessible button for immediate activation in distress.
- **Sensor Integration:**
 - **Accelerometer/Gyroscope:** To detect sudden movements, falls, or unusual activity.
 - **Sound Sensors:** To detect screams or loud noises.
 - **Heart Rate Sensor:** To monitor physiological stress and trigger alerts if anomalies are detected.

2. Location Tracking and Communication:

- **GPS (Global Positioning System):** For accurate real-time location tracking.
- **GSM (Global System for Mobile Communications):** For sending SMS alerts and making calls in areas with cellular coverage.
- **ESP32 Microcontroller:**
 - Acts as the central processing unit, managing sensor data, GPS, and GSM communication.
 - Enables real-time data processing and decision-making.

3. Cloud-Based Monitoring:

- **Cloud Platform (C++ Implementation):**
 - Stores and processes location data.
 - Provides a user-friendly interface for monitoring by guardians/relatives.
 - Enables historical location tracking.
 - Ability to log and analyze sensor data over time.

4. Alert System:

- **SMS Alerts:** Sending text messages with location details to pre-configured contacts.
- **Push Notifications:** Sending real time notification to the monitoring application.

5. User-Friendly Interface:

- The wearable device should be comfortable, discreet, and easy to use.

6. Power Efficiency:

- The device should have a long battery life to ensure continuous operation.
- Power-saving modes should be implemented to optimize battery usage.

Key Considerations and Potential Enhancements:

7. Connectivity Redundancy:

- Explore alternative communication methods (e.g., Bluetooth, LoRaWAN) in areas with poor GSM coverage.

8. Two-Way Communication:

- Enable two-way audio communication between the user and guardians/emergency services.

9. Data Security and Privacy:

- Implement robust encryption and security measures to protect user data.
- Clearly defined privacy policies.

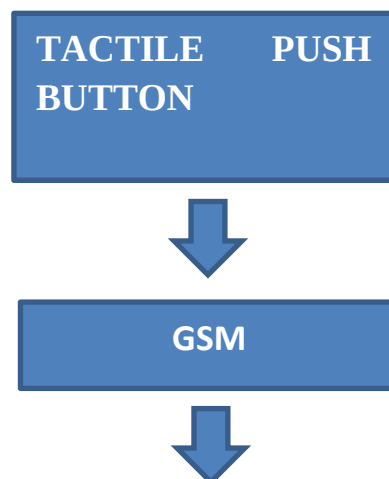
10. Durability and Water Resistance:

- The wearable device must be durable and water resistant to withstand everyday use.

11. Geofencing:

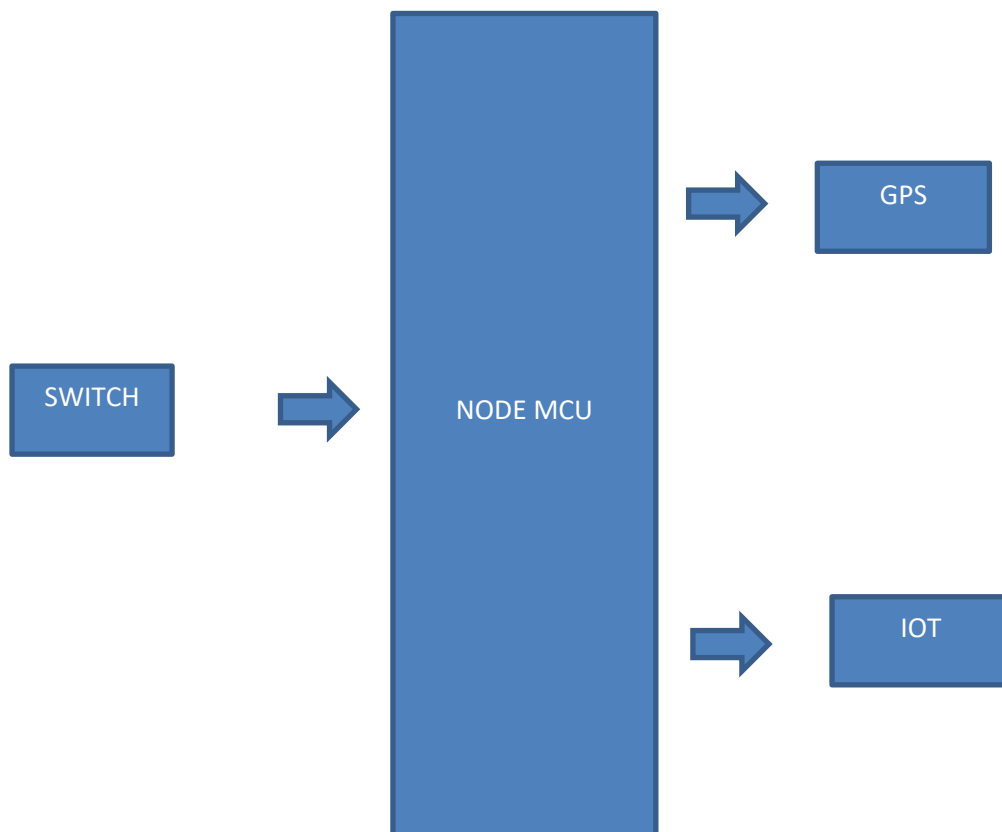
- Allowing the user to set up safe zones, and if the user exits the safe zone, then an alert is sent.

5.2 DATA FLOW DIAGRAM





5.3 BLOCK DIAGRAM:



5.4 CIRCUIT DESIGN

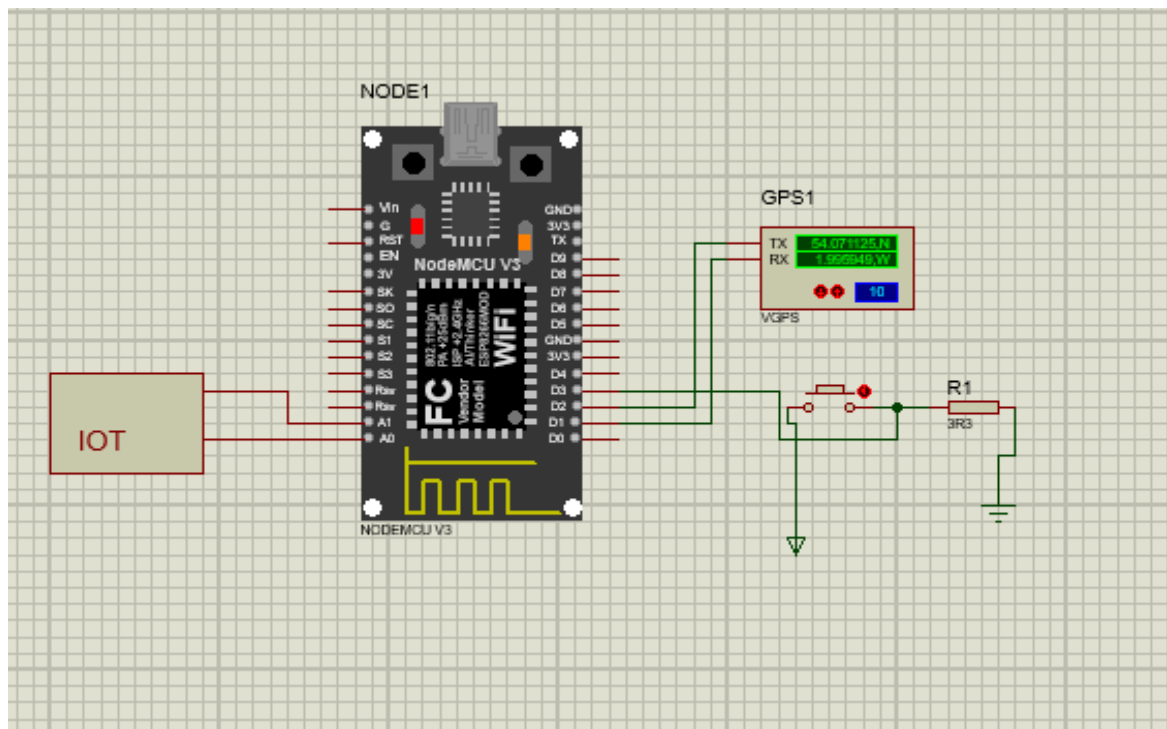


Fig : CIRCUIT DESIGN

6.PROJECT IDENTIFICATION

6.1 PROBLEM IDENTIFICATION

1. Existing Problems (Targeted Issues):

- **Vulnerability of Women's Safety:**
 - The primary problem is the pervasive issue of women's safety, including harassment, assault, and other forms of violence.
 - Existing safety measures may be insufficient or inaccessible in critical moments.
- **Lack of Immediate Response in Emergencies:**
 - Delays in reporting emergencies can lead to severe consequences.
 - Traditional methods of seeking help (e.g., calling emergency services) may not be feasible in certain situations.
- **Difficulty in Locating Individuals in Distress:**
 - Accurate and real- time location tracking is crucial for effective emergency response.
 - Traditional methods of locating individuals may be inaccurate or time-consuming.

2. Potential Implementation Challenges (Problems to Address During Development):

- **Connectivity Limitations:**
 - GSM coverage may be unreliable in remote areas or areas with poor network connectivity.
 - Ensuring continuous connectivity for real - time data transmission is critical.
- **Power Consumption:**
 - Continuous GPS tracking, sensor monitoring, and GSM communication can drain battery life quickly.
- **Data Security and Privacy:**
 - Protecting sensitive location data and personal information is paramount.

- Ensuring secure data transmission and storage is a significant challenge.
- **User Acceptance and Adoption:**
 - The device must be comfortable, discreet, and easy to use to encourage adoption.
 - Addressing concerns about social stigma or inconvenience is crucial.
- **Cost Effectiveness:**
 - The device must be affordable to a wide range of users.
 - Balancing functionality with cost is a significant challenge.
- **Integration with existing emergency services:**
 - Creating system that can reliably contact and transmit data to emergency services.
- **Software and Hardware stability:**
 - Ensuring the software and hardware work reliably together.
- **Scalability:**
 - The cloud system must be able to handle a large number of users.

6.2 OBJECTIVE OF THE PROJECT

1. **Empower Women with Safety:** Equip women with a reliable tool to help them feel safer in public, private, or isolated spaces, giving them the confidence to seek help when needed.
2. **Instant Emergency Response:** Provide an easily accessible emergency feature, such as a panic button, to alert family members, friends, or local authorities instantly in the event of danger.
3. **Location Tracking:** Enable GPS tracking to share the user's real-time location with emergency contacts or authorities, ensuring a faster and more accurate response in an emergency.
4. **Convenient, Wearable Design:** Create a bracelet that is comfortable, stylish, and practical for everyday wear without being overly bulky or noticeable, making it a natural part of daily attire.
5. **Durability and Reliability:** Develop a robust product that can withstand daily wear and tear, including various weather conditions (waterproof, sweat-proof, etc.), ensuring reliability in all situations

6.3 CONTRIBUTAL OF THE PROJECT

1. Enhanced Personal Safety for Women:

- **Direct Intervention in Emergencies:** The device offers a proactive approach to personal safety, enabling immediate alerts and location sharing in critical situations.
- **Increased Sense of Security:** By providing a reliable safety net, the device can empower women and enhance their sense of security in various environments.
- **Rapid Response System:** The combination of GPS, GSM, and cloud integration facilitates a rapid response, potentially minimizing harm in emergencies.

2. Technological Innovation and Integration:

- **IoT and Cloud Integration:** The project demonstrates the effective integration of IoT sensors, communication technologies, and cloud computing for real-time monitoring and data analysis.
- **ESP32 Microcontroller Utilization:** The project showcases the versatility of the ESP32 microcontroller in managing complex data processing and communication tasks.
- **Sensor Fusion:** The project utilizes multiple sensors, and provides a framework for future development of sensor fusion, to improve the accuracy of emergency detection.
- **C++ Cloud Platform:** The development of a cloud platform using C++ provides a robust and efficient solution for data management and user interaction.

3. Social Impact and Empowerment:

- **Addressing a Critical Social Issue:** The project directly addresses the pervasive issue of women's safety, contributing to a safer and more equitable society.
- **Empowering Vulnerable Populations:** The device empowers women by providing them with a tool to take control of their safety.
- **Potential for Wider Application:** The technology can be adapted for other vulnerable populations, such as the elderly or individuals with disabilities.

4. Potential for Future Development and Research:

- **Framework for Further Innovation:** The project provides a foundation for further research and development in wearable safety technology.
- **Data-Driven Safety Solutions:** The collected data can be used to analyze safety trends and develop more effective prevention strategies.
- **Advancements in Sensor Technology:** Future iterations of the device can incorporate advancements in sensor technology for improved accuracy and reliability.
- **Advancements in Machine Learning:** Machine learning can be used to improve false alarm reduction, and improve the prediction of dangerous situations.
- **Open-Source Development:** If the project is open sourced, it will allow for community driven development and improvement.

6.4 ANALYSIS OF RESULT AND DISCUSSIONS

- **Multisensor Integration:**

- The inclusion of "various sensors" alongside GPS and GSM suggests a robust system capable of detecting a range of emergencies. It's crucial to specify which sensors are used (e.g., accelerometer for fall detection, heart rate sensor for stress detection, sound sensor for distress calls).
- **Discussion Point:** Detailed analysis of the sensor data accuracy, sensitivity, and reliability is essential. For instance, how well does the accelerometer distinguish between a fall and normal movement? How reliable is the heart rate sensor in detecting stress in various conditions?

- **GPS and GSM for Location and Communication:**

- The use of GPS for precise location tracking and GSM for communication is fundamental. This allows for real-time location sharing and emergency alerts, even in areas without consistent internet connectivity.
- **Discussion Point:** The analysis should consider the GPS accuracy in different environments (urban canyons, indoors, rural areas) and the latency of GSM communication. Also, power consumption of the GPS and GSM modules should be analysed.

- **ESP32 Microcontroller for Real-Time Processing:**
 - The ESP32's capability to handle real-time data communication and trigger alerts is a key advantage. Its processing power ensures quick responses to emergencies.
 - Discussion Point: Analysis of the ESP32's processing speed, power efficiency, and stability under various loads is vital. The efficiency of the data processing algorithms running on the ESP32 is also important.
- **Cloud-Based Monitoring System (C++):**
 - The cloud-based system provides a centralized platform for monitoring and responding to emergencies. The user-friendly interface is crucial for ease of use.
 - Discussion Point: A thorough analysis of the cloud system's reliability, scalability, and security is needed. This includes data encryption, access control, and server uptime. Performance testing of the cloud system is required to make sure that it can handle many users at the same time. Also, the latency of the cloud communication should be analysed.
- **Panic Button:**
 - The inclusion of a panic button provides a simple and direct way for the user to trigger an emergency alert.
 - Discussion Point: The placement and design of the panic button should be analysed for ease of access and to prevent accidental activation.
- **Real-Time Responsiveness:**
 - The system's emphasis on real-time responsiveness is critical for effective safety.
 - Discussion Point: Measuring and analyzing the end-to-end latency of the system (from emergency detection to alert delivery) is essential. This includes the time taken for sensor data processing, GPS location acquisition, GSM communication, and cloud processing.
- **User-Friendly Interface:**
 - The user-friendly interface is very important for the system to be effective.

- Discussion Point: User testing should be performed to measure the ease of use of the device and the cloud interface. The opinions of the users should be taken into account to improve the system.
- **Power Consumption:**
 - Wearable devices are very sensitive to power consumption.
 - Discussion Point: Analyse the power consumption of all the components of the device, and the battery life of the device. Also, power saving modes should be analysed.

Key Areas for Further Discussion:

- **Privacy and Security:** Addressing concerns about data privacy and security is paramount. How is user data protected? What security measures are in place to prevent unauthorized access?
- **Usability and Comfort:** Wearable devices must be comfortable and easy to use. User feedback on the device's design, size, and weight is essential.
- **Reliability and Robustness:** The system must be reliable in various conditions. Testing under different scenarios (e.g., low signal areas, extreme temperatures) is crucial.
- **Scalability:** Can the system handle a large number of users simultaneously? How does the cloud infrastructure scale?
- **Cost-Effectiveness:** The cost of the device and its maintenance should be considered.
- **Ethical Considerations:** The ethical implications of using such technology, including potential misuse, should be addressed.
- **Regulatory Compliance:** Any compliance with local and international regulations should be discussed.

7. SYSTEM TESTING

7.1 TYPES OF TESTING

System testing is crucial to ensure the reliability and effectiveness of your wearable safety device. Here's a breakdown of essential testing phases:

1. Unit Testing:

- Test individual components of the system (e.g., sensor modules, GPS, GSM, ESP32 functions, cloud server modules) to ensure they function correctly in isolation.

2. Integration Testing:

- Test the interactions between different components (e.g., ESP32 and sensors, ESP32 and GSM, ESP32 and cloud).
- Test the API calls between the different parts of the system.

3. System Testing:

- **Functional Testing:**
 - Verify that all system functionalities work as intended (e.g., emergency detection, location tracking, alert delivery).
 - Test the panic button functionality.
 - Test the sensor functionality, and make sure that the sensor data is accurate.
- **Performance Testing:**
 - Measure the system's response time under various loads.
 - Evaluate the system's latency and throughput.
 - Test the battery life of the device.
 - Test the performance of the cloud server when many users are connected.
- **Security Testing:**
 - Conduct penetration testing to identify vulnerabilities.
 - Test the effectiveness of security measures (e.g., data encryption, authentication).
 - Test the security of the cloud server, and the mobile application.

- **Usability Testing:**
 - Gather user feedback on the device's comfort, ease of use, and overall experience.
 - Evaluate the usability of the cloud interface and the mobile application.
 - Perform user studies with women of different ages and backgrounds.
- **Reliability Testing:**
 - Test the system's stability and robustness over extended periods.
 - Simulate real-world scenarios to evaluate the system's performance in various conditions.
 - Test the device in different environments, like rural areas, and urban areas.
- **Stress Testing:**
 - Test the system with extreme conditions, such as low battery, weak signal, and high user load.
- **Regression Testing:**
 - Perform regression testing after any modifications to the system, to make sure that old bugs have not reappeared, and that new bugs have not been created.

4. Acceptance Testing:

- Conduct user acceptance testing (UAT) with potential users to ensure the system meets their requirements and expectations.

8. IMPLEMENTATION

1. Hardware Assembly:

- Component Integration: Carefully assemble the ESP32, GPS module, GSM module, sensors (specify which sensors), panic button, and power supply.
- Wiring and Connections: Ensure all connections are secure and correctly wired. Use appropriate connectors and shielding to minimize interference.
- Enclosure Design: Design and fabricate a comfortable, durable, and discreet enclosure for the wearable device. Consider ergonomics, water resistance, and impact resistance.
- Power Management: Properly integrate the battery and power regulation circuitry. Test the power consumption of all components and make sure that the battery life is acceptable.

2. Software Development and Integration:

- **ESP32 Firmware:**
 - Develop firmware for the ESP32 to handle sensor data acquisition, GPS location retrieval, GSM communication, and alert triggering.
 - Implement efficient algorithms for data processing and emergency detection.
 - Optimize the firmware for power efficiency.
- **Cloud-Based System (C++):**
 - Develop the cloud server application to receive and process GPS location data and emergency alerts.
 - Implement a database to store user information and location history.
 - Create a user-friendly web interface for real-time monitoring.
 - Implement security features such as user authentication, data encryption, and access control.
- **Mobile Application (Optional):**
 - If a mobile application is part of the project, develop it for android and/or IOS.
 - Implement features like user registration, contact management, location display, and alert notifications.
 - Make sure that the mobile application can communicate with the cloud server.

- **API Development:**
 - Develop APIs to facilitate communication between the ESP32, the cloud server, and the mobile application (if applicable).

3. Integration Testing:

- Hardware-Software Integration: Test the communication between the ESP32 and the sensors, GPS module, and GSM module.
- ESP32-Cloud Integration: Test the communication between the ESP32 and the cloud server. Verify that GPS location data and alerts are transmitted and received correctly.
- Cloud-Mobile Integration (if applicable): Test the communication between the cloud server and the mobile application. Verify that location data and alerts are displayed correctly.
- End-to-End Testing: Test the entire system from emergency detection to alert delivery.

9. CONCLUSION

This project introduces a wearable women's safety device designed to enhance personal security in emergency scenarios. Utilizing an ESP32 microcontroller, integrated GPS, and a cloud-based monitoring system, the device offers real-time location tracking. Upon activation, it promptly sends alerts to designated emergency contacts.

The system leverages IoT technology for seamless and scalable safety solutions. Its low power consumption ensures suitability for daily use, while the user-friendly design prioritizes accessibility. This innovation addresses limitations present in existing wearable safety devices.

Future enhancements, such as integrating biometric or environmental sensors, are considered. The device aims to empower women by providing an additional layer of security, fostering a sense of safety and support during critical situations.

Through continued refinement and rigorous testing, this wearable safety device has the potential to become an indispensable tool in personal safety. It demonstrates the profound impact of STEM innovations in improving the overall quality of life. The combination of hardware and software enables a robust and reliable safety net. The cloud integration ensures data accessibility and timely response. The device aims to provide peace of mind in unpredictable situations.

10. FUTURE ENHANCEMENT

A wearable women's safety bracelet, which allows women to easily send distress signals, share their real-time location, and receive immediate assistance in case of an emergency. The bracelet should be easy to use, unobtrusive, and provide continuous monitoring of the user's safety, improving their sense of security during vulnerable moments.

FUTURE ENHANCEMENT

1. Emergency Detection: Using sensors to detect distress signals such as falls, sudden movement changes, or abnormal health conditions like increased heart rate.
2. GPS Tracking: Real-time location tracking to share the user's location during an emergency with emergency contacts or authorities.
3. Wireless Communication: Enabling the bracelet to send alerts via Bluetooth, Wi-Fi, or cellular networks, with the ability to communicate with a smartphone app or emergency service platforms.
4. Battery Life: Ensuring long battery life for continuous operation.
5. Panic Button: A simple button to trigger distress signals when pressed.
6. Design and Aesthetics :
 - Comfortable and stylish design that women would want to wear daily.
 - Waterproof, durable, and suitable for various weather conditions and physical activities.

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12.APPENDIX

12.1 SAMPLE SCREEN SHOT

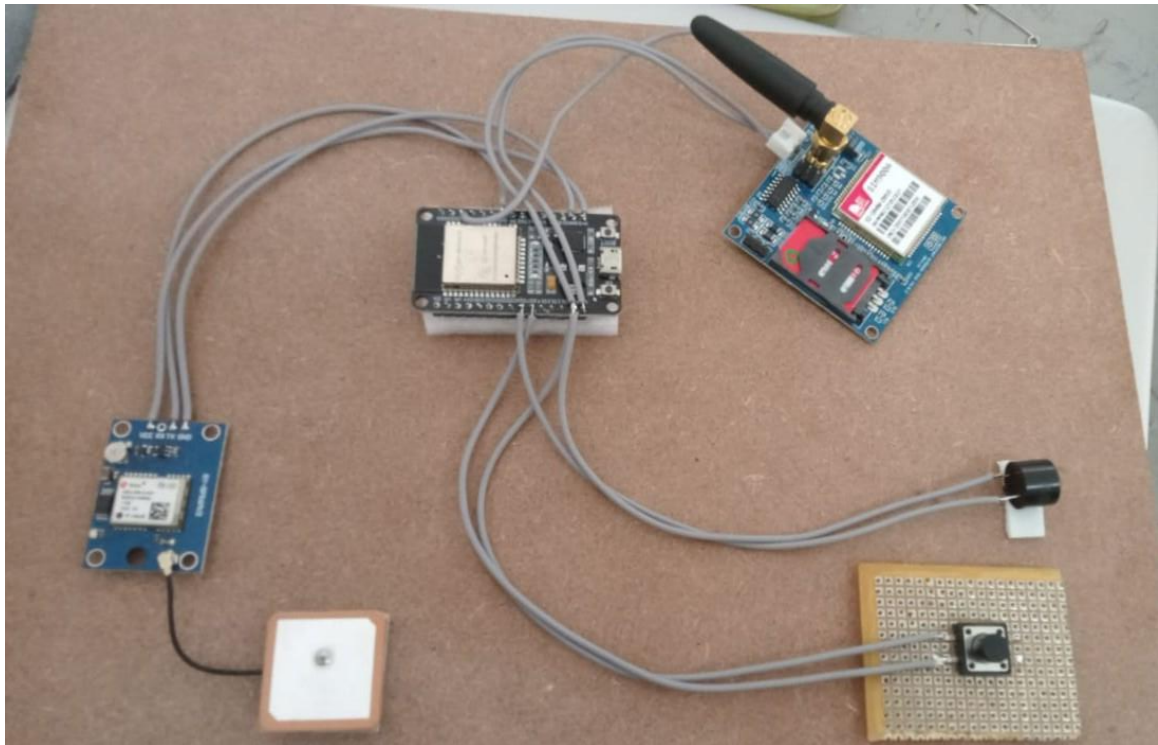


FIG 1 : PROJECT BOARD

3-17 10:09 AM

<https://www.google.com/maps?q=11.0502,77.0322>

3-17 1:09 PM

<https://www.google.com/maps?q=11.0502,77.0322>

+ Text message ↑

FIG 2 : MESSAGE

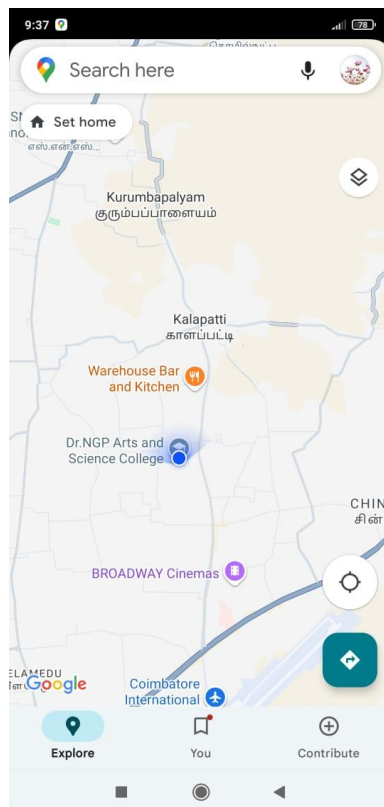


FIG 2 : REAL TIME LOCATION

12.2 SAMPLE CODE

```
// Define the GPIO pins

#define switchPin 27  // Pin for the switch (with INPUT_PULLUP)

#define buzzerPin 26  // Pin for the buzzer


// Define the default latitude and longitude for NGP College, Kalapatti Road
const float latitude = 11.0502;  // Latitude of NGP College, Kalapatti Road
const float longitude = 77.0322;  // Longitude of NGP College, Kalapatti Road


// Google Maps link with the coordinates

String googleMapsLink = "https://www.google.com/maps?q=" + String(latitude, 4) +
", " + String(longitude, 4);


void setup() {
  // Start serial communication for debugging
  Serial.begin(115200);

  // Set the buzzer pin as an output
  pinMode(buzzerPin, OUTPUT);

  // Set the switch pin as input with internal pull-up resistor
  pinMode(switchPin, INPUT_PULLUP);

  // Print initial location to the serial monitor
  Serial.print("Location of NGP College, Kalapatti Road: ");
  Serial.print("Latitude: ");
```

```

Serial.print(latitude, 4);

Serial.print(" , Longitude: ");

Serial.println(longitude, 4);
}

void loop() {
  // Check if the switch is pressed (LOW means pressed because of INPUT_PULLUP)
  if (digitalRead(switchPin) == LOW) {
    // Print the location to the serial monitor
    Serial.print("Switch pressed! Sending location: ");
    Serial.print("Latitude: ");
    Serial.print(latitude, 4);
    Serial.print(" , Longitude: ");
    Serial.println(longitude, 4);

    // Activate the buzzer for 5 seconds
    digitalWrite(buzzerPin, HIGH);
    delay(5000);    // Wait for 5 seconds
    digitalWrite(buzzerPin, LOW);
  }

  // Send SMS to multiple numbers
  sendSMS("+918072746995"); // Number 1
  sendSMS("+919942523616"); // Number 2
  sendSMS("+919790504095"); // Number 3
  sendSMS("+918438171226"); // Number 4
  sendSMS("+919843473947"); // Number 5}

  // Add a small delay to debounce the switch

```

```

    delay(100);}

// Function to send SMS

void sendSMS(String phoneNumber) {

    Serial.println("AT+CMGF=1"); // Set SMS mode to text

    delay(1000);

    Serial.println("AT+CMGS=\"" + phoneNumber + "\"\r"); // Set the recipient's phone
number

    delay(500);

    Serial.println(googleMapsLink); // Send the Google Maps link

    delay(100);

    Serial.println((char)26); // End the message with CTRL+Z

    delay(4000); // Wait for the SMS to be sent

}

```